Sri Lanka Institute of Information Technology

IT3010 - Network Design and Management

DHCP Server Configuration and Testing Using VMware

Network Design and Management
Lab Sheet 02

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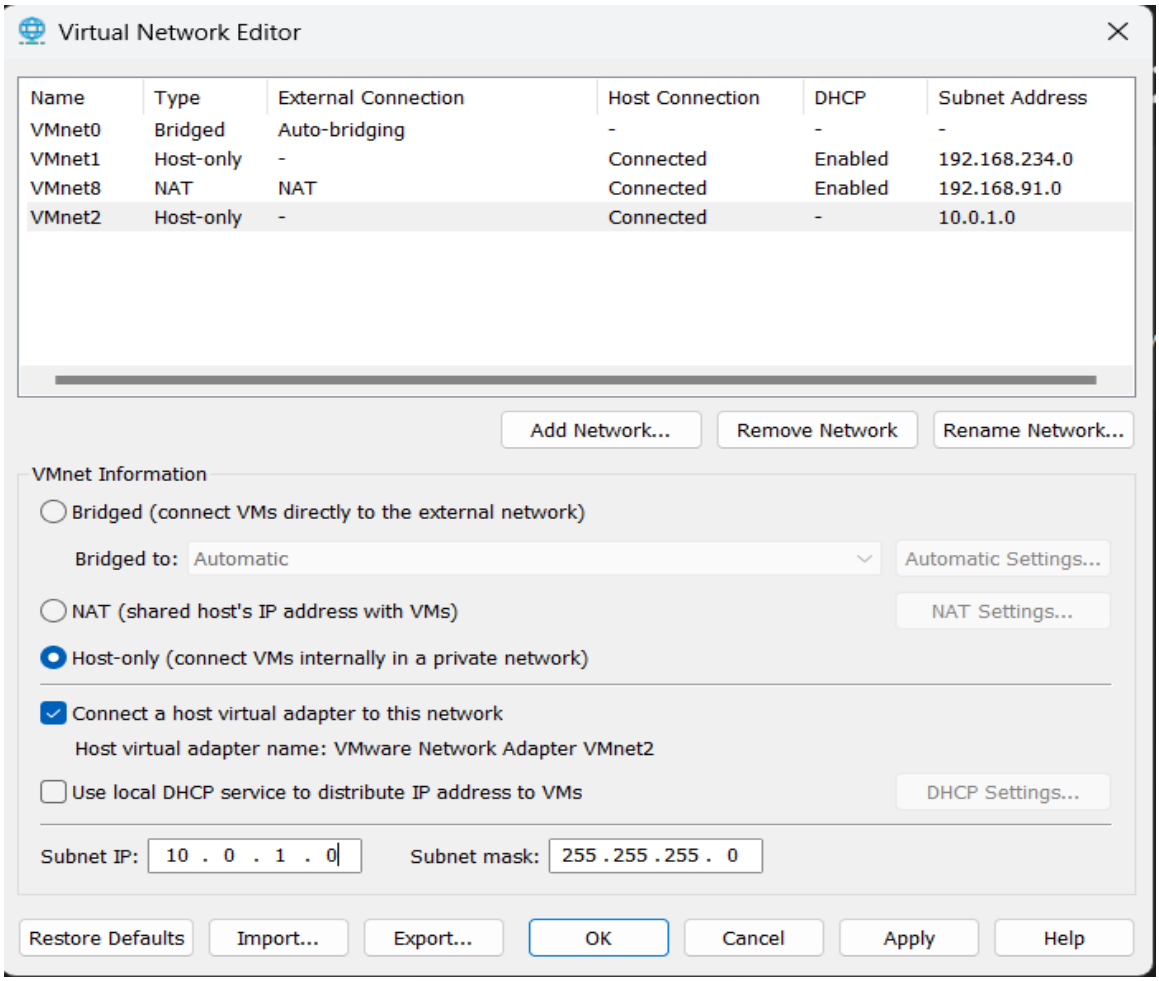
1. Objective

The objective of this lab is to install and configure a DHCP server on a CentOS virtual machine using VMware Workstation. The DHCP server is set up to automatically assign IP addresses and other network configuration settings to client machines within the same virtual network.

A Fedora client machine is configured to obtain its IP address dynamically from the DHCP server. Successful IP assignment and communication between the server and client confirm that the DHCP service is functioning correctly. This practical demonstrates how DHCP simplifies network management by automatically distributing IP addresses, gateways, subnet masks, and DNS settings without manual configuration.

2. Step 1 – Disabling VMware DHCP Service in VMnet2

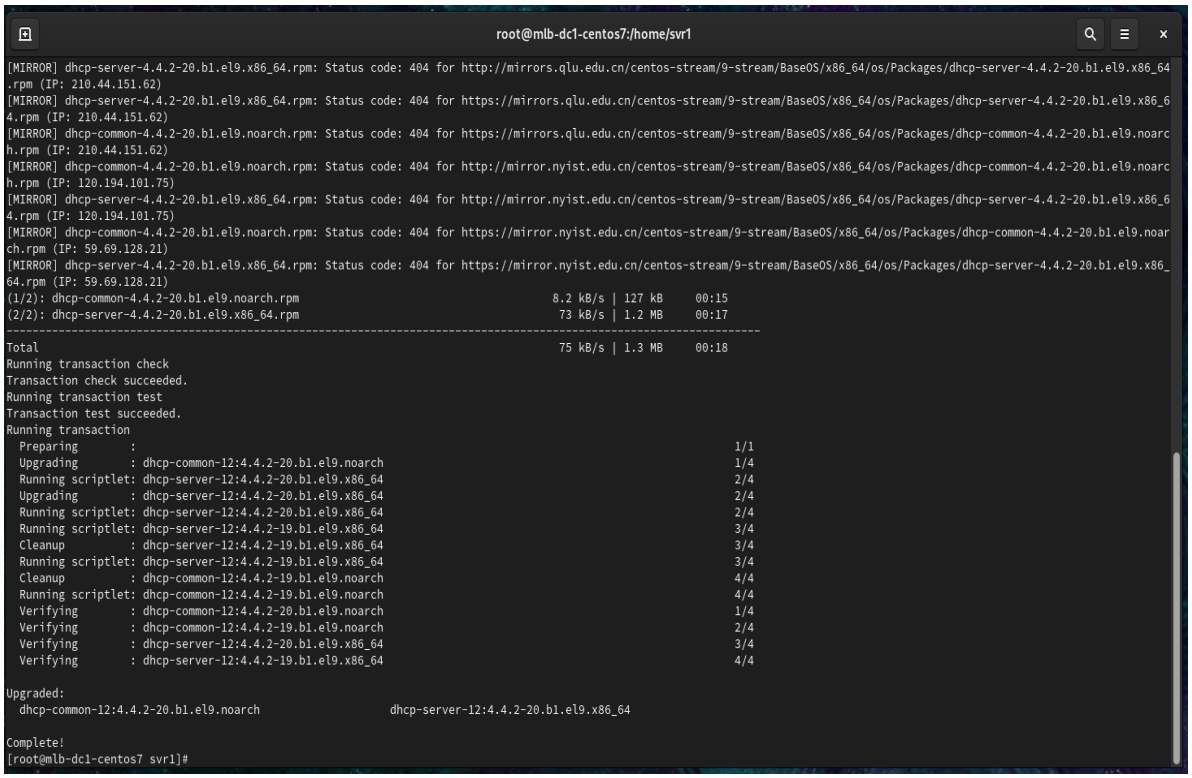
The default DHCP service for the VMnet2 virtual network was disabled using the VMware Virtual Network Editor. This prevents automatic IP assignment from VMware and allows the CentOS server to act as the primary DHCP server for the network, ensuring proper IP address management and avoiding conflicts.



3. Step 2 - DHCP Installation

The DHCP server package was installed on the CentOS 7 virtual machine using the command: sudo dnf install dhcp-server –y

This installed all necessary files and services required to run the DHCP server, enabling the server to dynamically assign IP addresses and network settings to client machines on the virtual network.



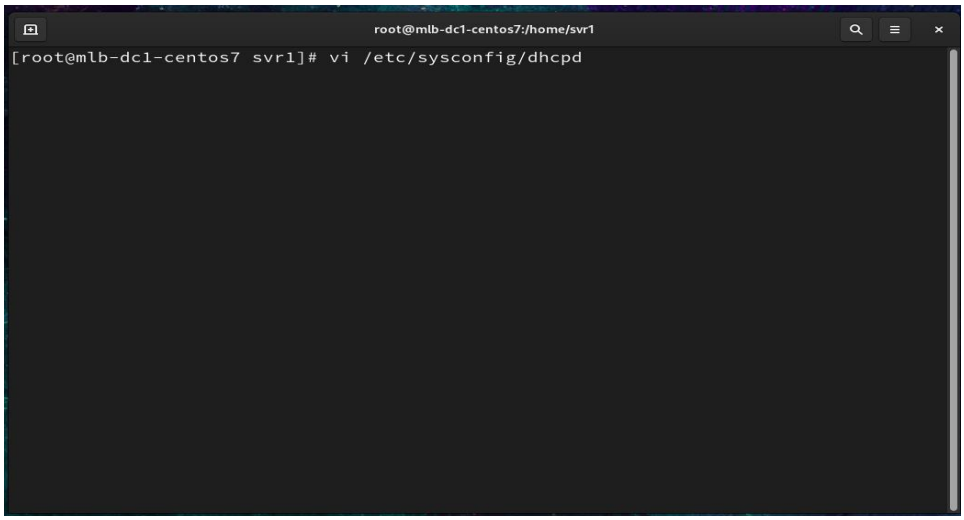
```
root@mlb-dc1-centos7:/home/svr1
[MIRROR] dhcp-server-4.4.2-20.b1.el9.x86_64.rpm: Status code: 404 for http://mirrors.qlu.edu.cn/centos-stream/9-stream/BaseOS/x86_64/os/Packages/dhcp-server-4.4.2-20.b1.el9.x86_64
.rpm (IP: 210.44.151.62)
[MIRROR] dhcp-server-4.4.2-20.b1.el9.x86_64.rpm: Status code: 404 for https://mirrors.qlu.edu.cn/centos-stream/9-stream/BaseOS/x86_64/os/Packages/dhcp-server-4.4.2-20.b1.el9.x86_6
4.rpm (IP: 210.44.151.62)
[MIRROR] dhcp-common-4.4.2-20.b1.el9.noarch.rpm: Status code: 404 for https://mirrors.qlu.edu.cn/centos-stream/9-stream/BaseOS/x86_64/os/Packages/dhcp-common-4.4.2-20.b1.el9.noarc
h.rpm (IP: 210.44.151.62)
[MIRROR] dhcp-common-4.4.2-20.b1.el9.noarch.rpm: Status code: 404 for http://mirror.nyist.edu.cn/centos-stream/9-stream/BaseOS/x86_64/os/Packages/dhcp-common-4.4.2-20.b1.el9.noarc
h.rpm (IP: 120.194.101.75)
[MIRROR] dhcp-server-4.4.2-20.b1.el9.x86_64.rpm: Status code: 404 for http://mirror.nyist.edu.cn/centos-stream/9-stream/BaseOS/x86_64/os/Packages/dhcp-server-4.4.2-20.b1.el9.x86_6
4.rpm (IP: 120.194.101.75)
[MIRROR] dhcp-common-4.4.2-20.b1.el9.noarch.rpm: Status code: 404 for https://mirror.nyist.edu.cn/centos-stream/9-stream/BaseOS/x86_64/os/Packages/dhcp-common-4.4.2-20.b1.el9.noar
ch.rpm (IP: 59.69.128.21)
[MIRROR] dhcp-server-4.4.2-20.b1.el9.x86_64.rpm: Status code: 404 for https://mirror.nyist.edu.cn/centos-stream/9-stream/BaseOS/x86_64/os/Packages/dhcp-server-4.4.2-20.b1.el9.x86_
64.rpm (IP: 59.69.128.21)
(1/2): dhcp-common-4.4.2-20.b1.el9.noarch.rpm                                   8.2 kB/s | 127 kB     00:15
(2/2): dhcp-server-4.4.2-20.b1.el9.x86_64.rpm                                    73 kB/s | 1.2 MB     00:17
--------------------------------------------------------------------------------------------------------------
Total                                                                            75 kB/s | 1.3 MB     00:18
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                      1/1
  Upgrading        : dhcp-common-12:4.4.2-20.b1.el9.noarch                                                1/4
  Running scriptlet: dhcp-server-12:4.4.2-20.b1.el9.x86_64                                                2/4
  Upgrading        : dhcp-server-12:4.4.2-20.b1.el9.x86_64                                                2/4
  Running scriptlet: dhcp-server-12:4.4.2-20.b1.el9.x86_64                                                2/4
  Running scriptlet: dhcp-server-12:4.4.2-19.b1.el9.x86_64                                                3/4
  Cleanup          : dhcp-server-12:4.4.2-19.b1.el9.x86_64                                                3/4
  Running scriptlet: dhcp-server-12:4.4.2-19.b1.el9.x86_64                                                3/4
  Cleanup          : dhcp-common-12:4.4.2-19.b1.el9.noarch                                                4/4
  Running scriptlet: dhcp-common-12:4.4.2-19.b1.el9.noarch                                                4/4
  Verifying        : dhcp-common-12:4.4.2-20.b1.el9.noarch                                                1/4
  Verifying        : dhcp-common-12:4.4.2-19.b1.el9.noarch                                                2/4
  Verifying        : dhcp-server-12:4.4.2-20.b1.el9.x86_64                                                3/4
  Verifying        : dhcp-server-12:4.4.2-19.b1.el9.x86_64                                                4/4

Upgraded:
  dhcp-common-12:4.4.2-20.b1.el9.noarch                    dhcp-server-12:4.4.2-20.b1.el9.x86_64

Complete!
[root@mlb-dc1-centos7 svr1]#
```

4. Step 3 - DHCP Configuration

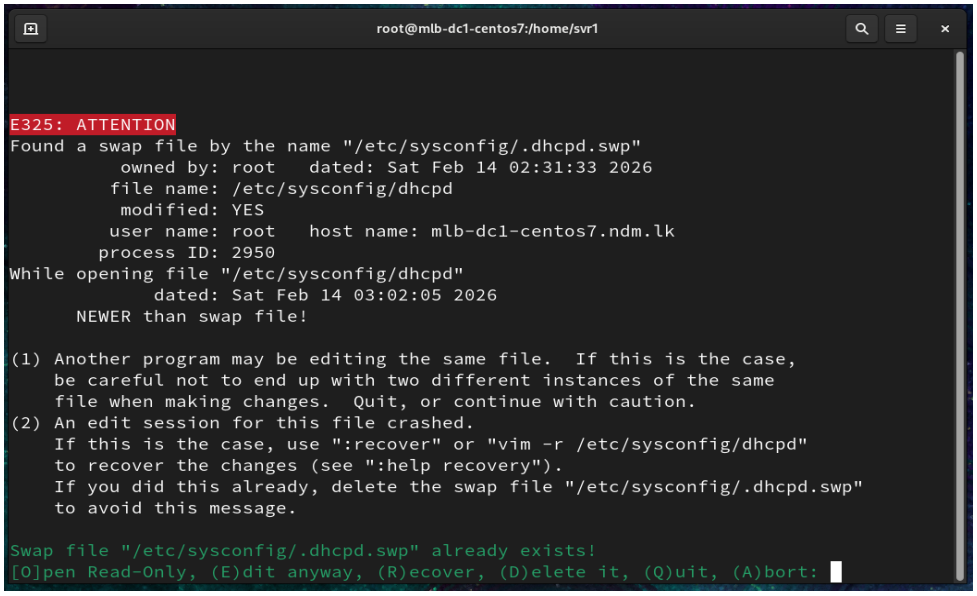
After installing the DHCP server, we must configure it so that it can assign IP addresses to client machines automatically.

This configuration is done by editing the main DHCP configuration file:

/etc/dhcp/dhcpd.conf



Go to file directory press “e” and go to the edit mode enter the “I”



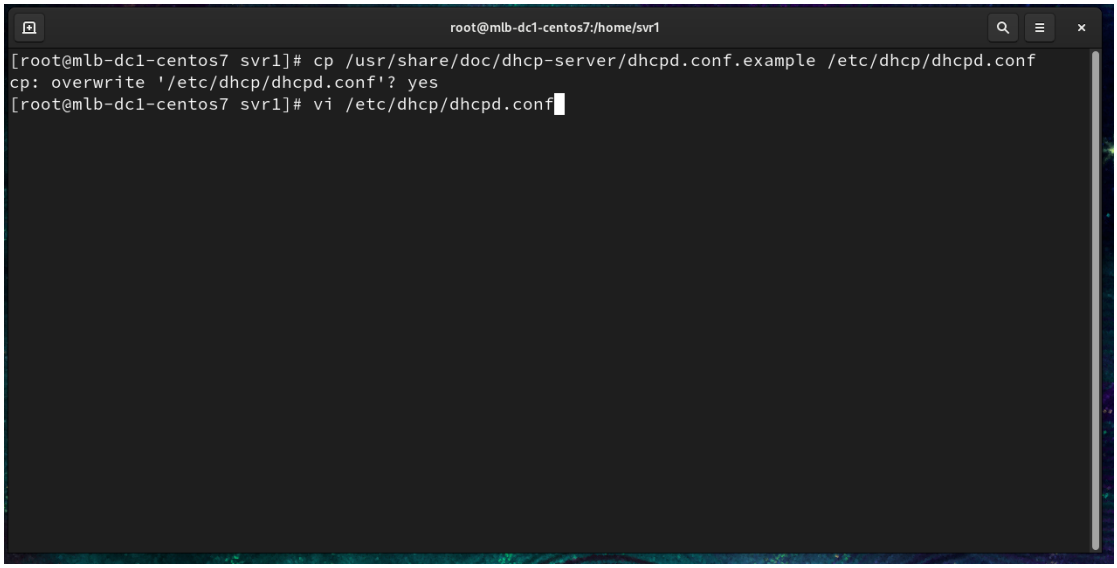
Within the edit mode enter the “DHCPARGS =ens36”

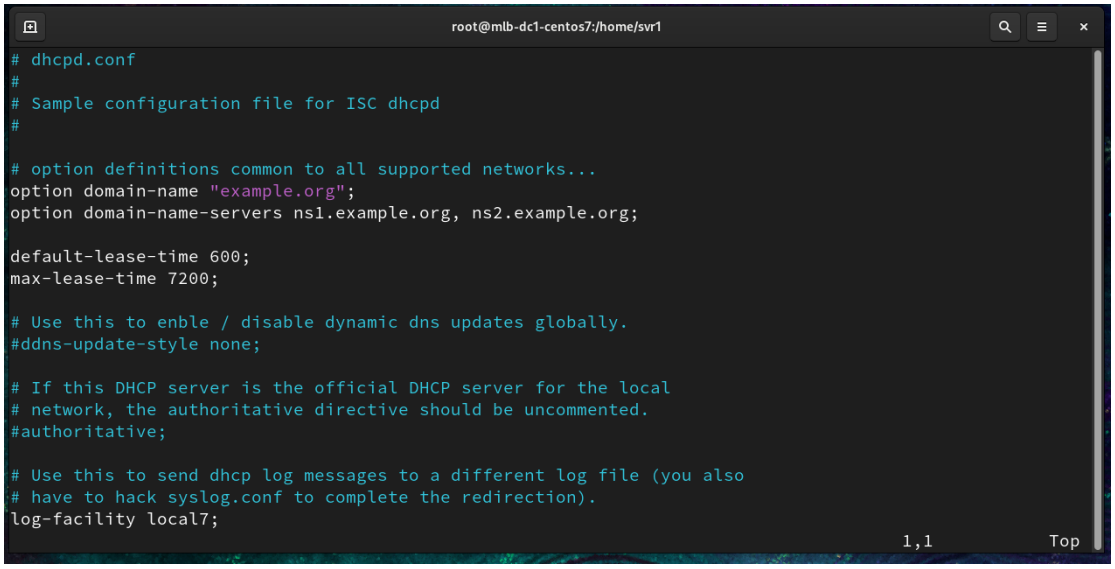
```
root@mlb-dc1-centos7:/home/svr1
# WARNING: This file is NOT used anymore.

# If you are here to restrict what interfaces should dhcpd listen on,
# be aware that dhcpd listens *only* on interfaces for which it finds subnet
# declaration in dhcpd.conf. It means that explicitly enumerating interfaces
# also on command line should not be required in most cases.

# If you still insist on adding some command line options,
# copy dhcpd.service from /lib/systemd/system to /etc/systemd/system and modify
# it there.
# https://fedoraproject.org/wiki/Systemd#How_do_I_customize_a_unit_file.2F_add_a_cust
om_unit_file.3F

# example:
# $ cp /usr/lib/systemd/system/dhcpd.service /etc/systemd/system/
# $ vi /etc/systemd/system/dhcpd.service
# $ ExecStart=/usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --
no-pid <your_interface_name(s)>
# $ systemctl --system daemon-reload
# $ systemctl restart dhcpd.service
DHCPDARGS=ens36
~
~
~
-- INSERT --                                                    19,16         All
```

This ‘ens36’ is network adapter of configuration. That can be see using “ifconfig” command

```
root@mlb-dc1-centos7:/home/svr1
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qle
n 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group de
fault qlen 1000
    link/ether 00:0c:29:2b:40:13 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.234.128/24 brd 192.168.234.255 scope global dynamic noprefixroute en
s33
       valid_lft 1659sec preferred_lft 1659sec
    inet6 fe80::20c:29ff:fe2b:4013/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: ens36: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group de
fault qlen 1000
    link/ether 00:0c:29:2b:40:1d brd ff:ff:ff:ff:ff:ff
    altname enp2s4
    inet 10.0.1.2/24 brd 10.0.1.255 scope global noprefixroute ens36
       valid_lft forever preferred_lft forever
    inet6 fe80::1d81:51ca:cc03:340c/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@mlb-dc1-centos7 svr1]#
```

Client Testing Then go to the dhcp configuration file to /etc/dhcp/ directory and enter the "yes" Directory: cp /usr/share/doc/dhcp-server/dhcpd.conf.example /etc/dhcp/dhcpd.conf





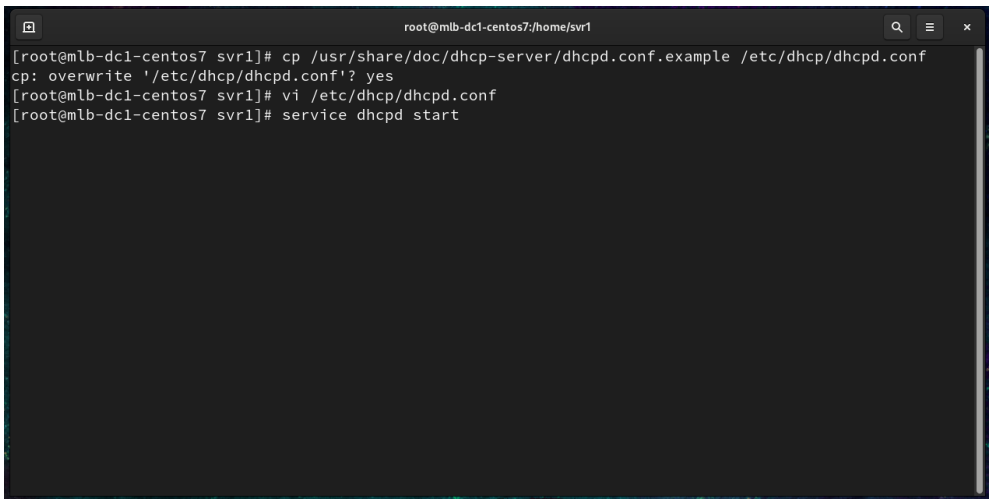
Press “I” and Make the changes as shown below screenshots

```
root@mlb-dc1-centos7:/home/svr1
# dhcpd.conf
#
# Sample configuration file for ISC dhcpd
#

# option definitions common to all supported networks...
option domain-name "dsnm.sub";
option domain-name-servers server.unixmen.local;

default-lease-time 600;
max-lease-time 7200;

# Use this to enble / disable dynamic dns updates globally.
#ddns-update-style none;

# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
authoritative;

# Use this to send dhcp log messages to a different log file (you also
# have to hack syslog.conf to complete the redirection).
log-facility local7;
-- INSERT --                                                    1,1           Top
```

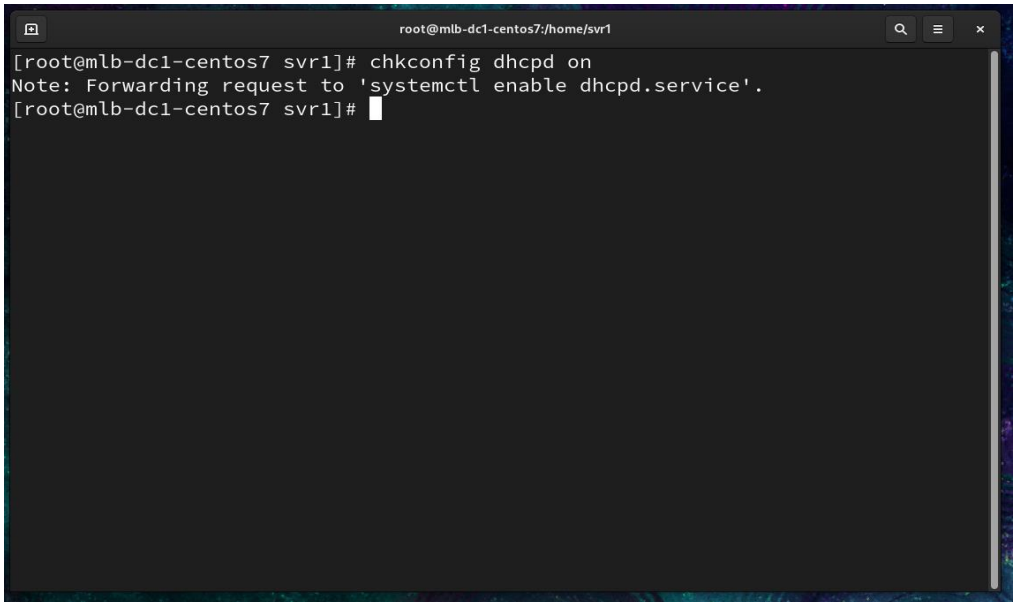
```
root@mlb-dc1-centos7:/home/svr1
  option routers rtr-239-0-1.example.org, rtr-239-0-2.example.org;
}

# This declaration allows BOOTP clients to get dynamic addresses,
# which we don't really recommend.

subnet 10.254.239.32 netmask 255.255.255.224 {
  range dynamic-bootp 10.254.239.40 10.254.239.60;
  option broadcast-address 10.254.239.31;
  option routers rtr-239-32-1.example.org;
}

# A slightly different configuration for an internal subnet.
subnet 10.0.1.0 netmask 255.255.255.0 {
  range 10.0.1.20 10.0.1.30;
  option domain-name-servers server.unixmen.local;
  option domain-name "dsnm.sub";
  option routers 10.0.1.1;
  option broadcast-address 10.0.1.255;
  default-lease-time 600;
  max-lease-time 7200;
}
-- INSERT --                                                   39,1          40%
```

5. Step 4 - Starting and Testing DHCP Service

After configuring the DHCP server, the service must be started so that it can begin assigning IP addresses to client machines. We also need to check whether the service is running properly and ensure that it starts automatically after a system reboot.



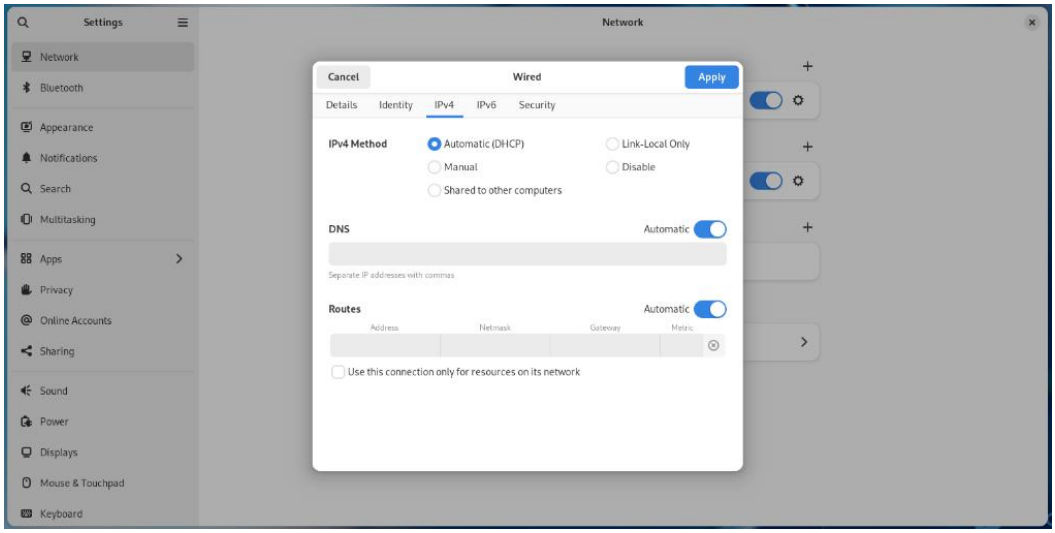
If you want to start up the DHCP server at logon to the server session use “chkconfig dhcpd on” Command



Checking the status of DHCP server and client, using “systemctl status dhcpd” Command

```
root@mlb-dc1-centos7:/home/svr1
● dhcpd.service - DHCPv4 Server Daemon
     Loaded: loaded (/usr/lib/systemd/system/dhcpd.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-02-16 14:50:54 EST; 24min ago
       Docs: man:dhcpd(8)
             man:dhcpd.conf(5)
   Main PID: 3015 (dhcpd)
     Status: "Dispatching packets..."
      Tasks: 1 (limit: 10717)
     Memory: 5.9M
        CPU: 65ms
     CGroup: /system.slice/dhcpd.service
             └─3015 /usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid ens36

Feb 16 14:50:54 mlb-dc1-centos7.ndm.lk dhcpd[3015]: Source compiled to use binary-leases
Feb 16 14:50:54 mlb-dc1-centos7.ndm.lk dhcpd[3015]: Wrote 0 class decls to leases file.
Feb 16 14:50:54 mlb-dc1-centos7.ndm.lk dhcpd[3015]: Wrote 0 deleted host decls to leases file.
Feb 16 14:50:54 mlb-dc1-centos7.ndm.lk dhcpd[3015]: Wrote 0 new dynamic host decls to leases file.
Feb 16 14:50:54 mlb-dc1-centos7.ndm.lk dhcpd[3015]: Wrote 0 leases to leases file.
Feb 16 14:50:54 mlb-dc1-centos7.ndm.lk dhcpd[3015]: Listening on LPF/ens36/00:0c:29:2b:40:1d/10.0.1.0/24
Feb 16 14:50:54 mlb-dc1-centos7.ndm.lk dhcpd[3015]: Sending on   LPF/ens36/00:0c:29:2b:40:1d/10.0.1.0/24
Feb 16 14:50:54 mlb-dc1-centos7.ndm.lk dhcpd[3015]: Sending on   Socket/fallback/fallback-net
Feb 16 14:50:54 mlb-dc1-centos7.ndm.lk dhcpd[3015]: Server starting service.
Feb 16 14:50:54 mlb-dc1-centos7.ndm.lk systemd[1]: Started DHCPv4 Server Daemon.
~
lines 1-23/23 (END)
```

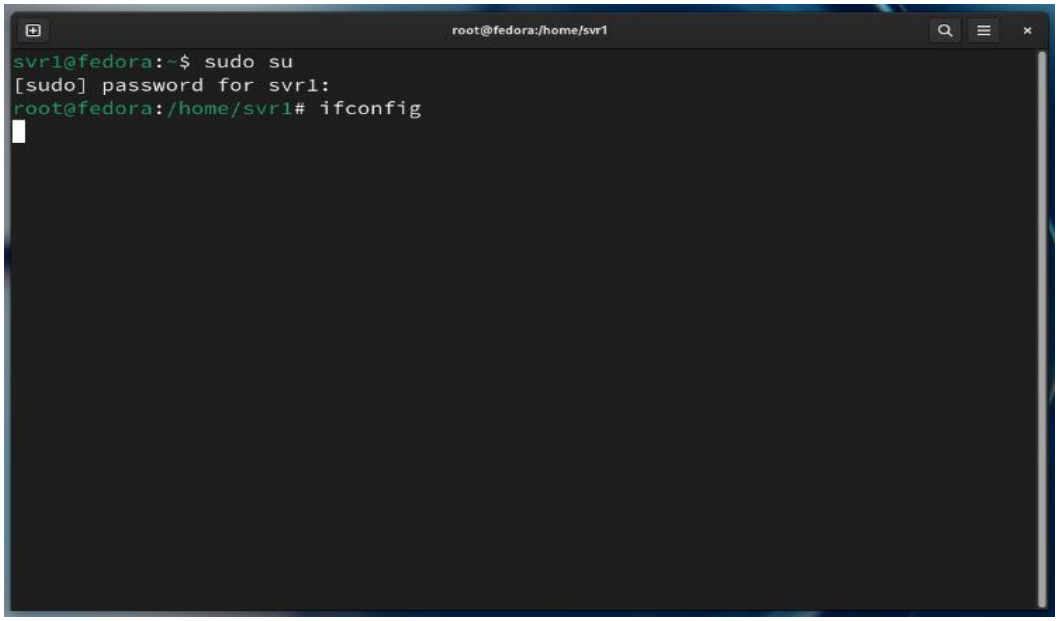
Then go to the client [FedoraOS] and switch the IPv4 as Automatic (DHCP). Then you can see it is changing “Connected Mode”

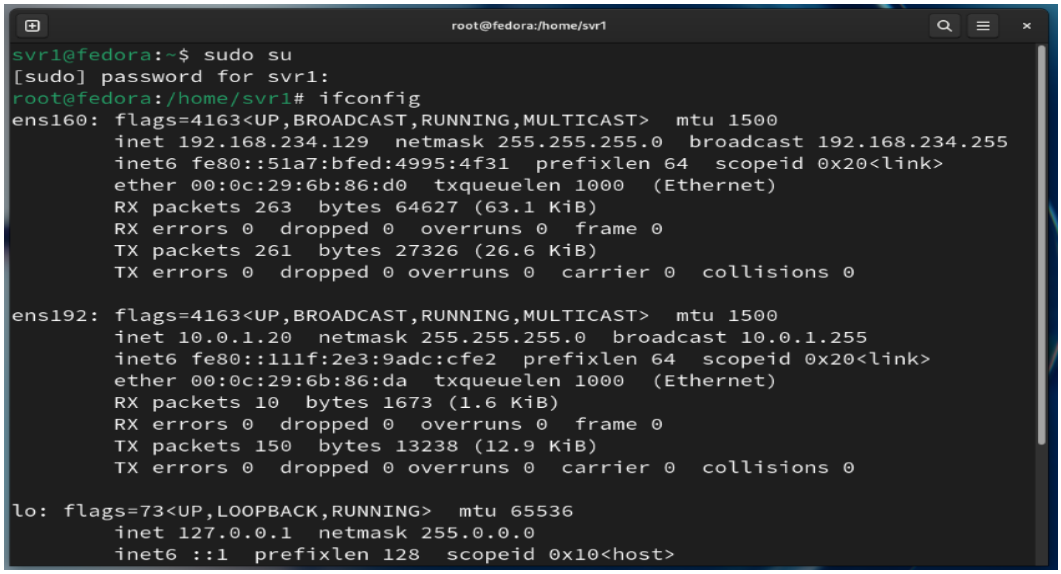


6. Self-Study

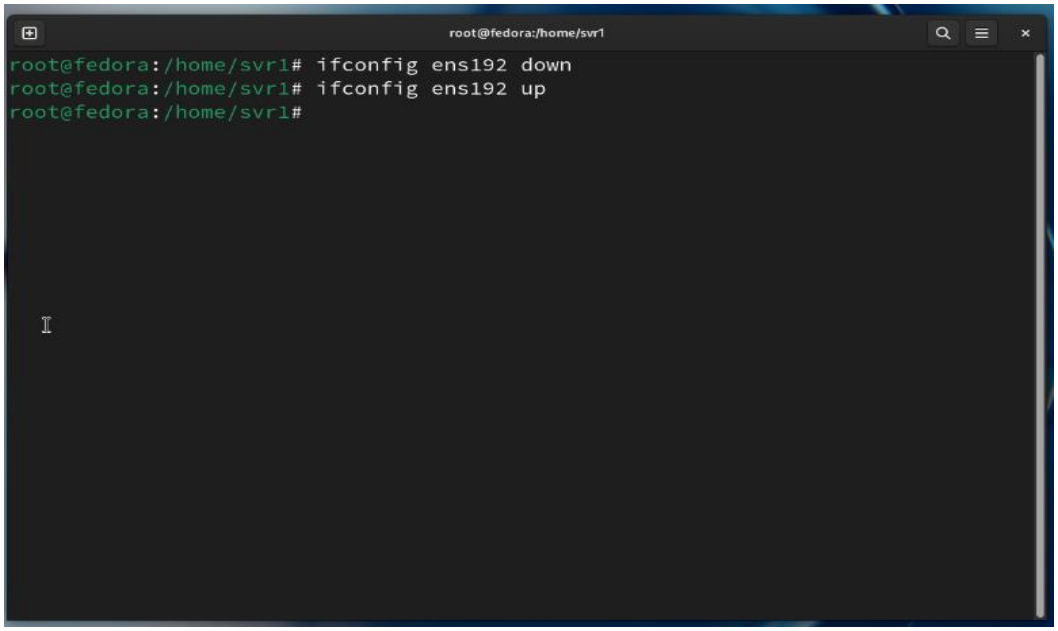
1. Changing IP Configurations in Linux

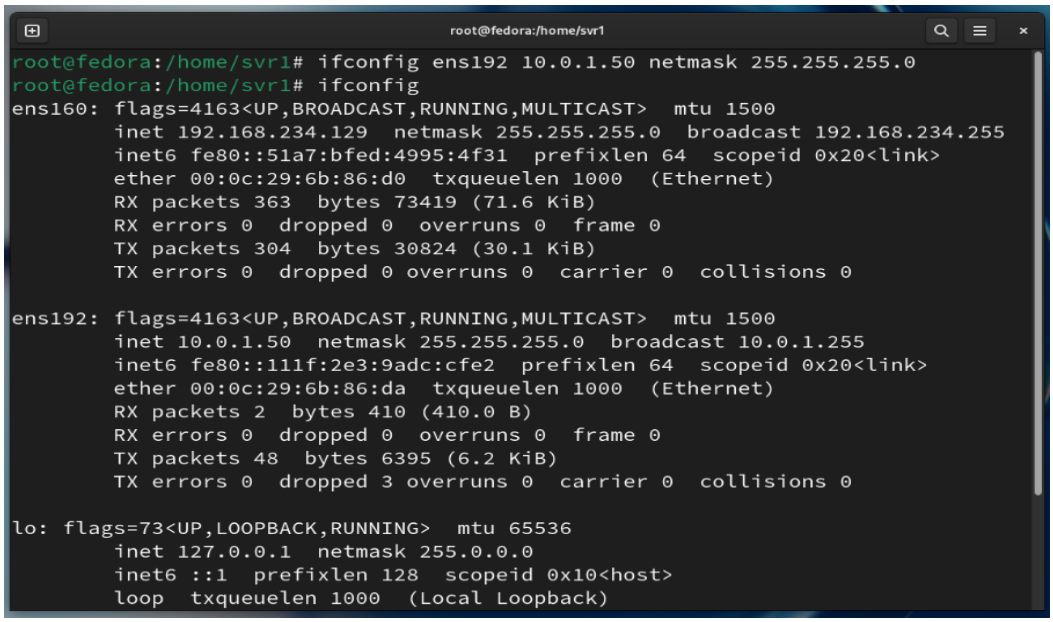
Go to the root using "sudo su" and identify the interface you need to configure using the ifconfig command.



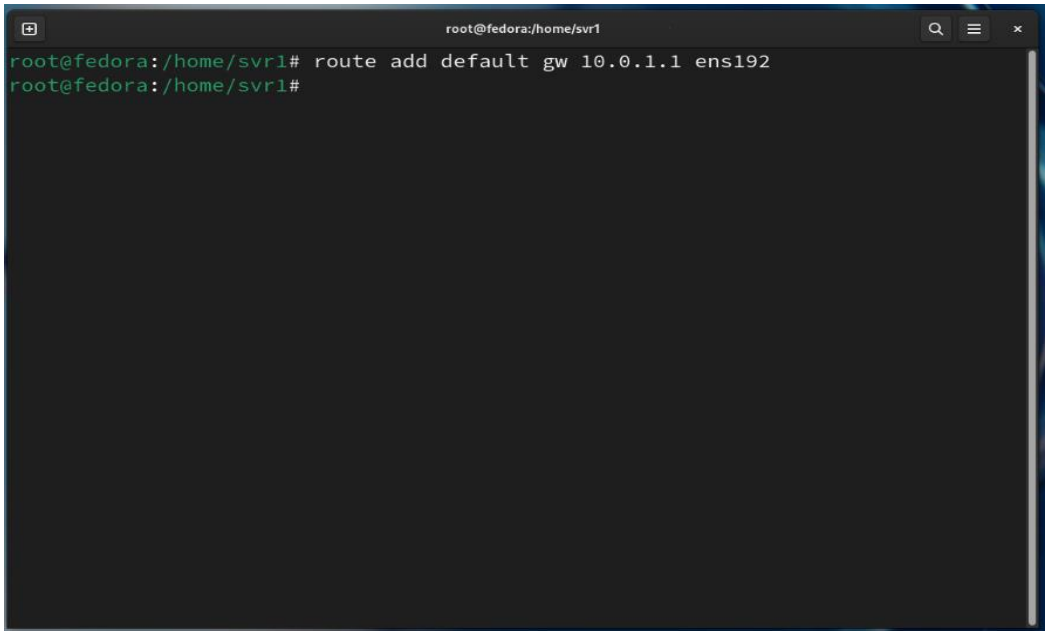


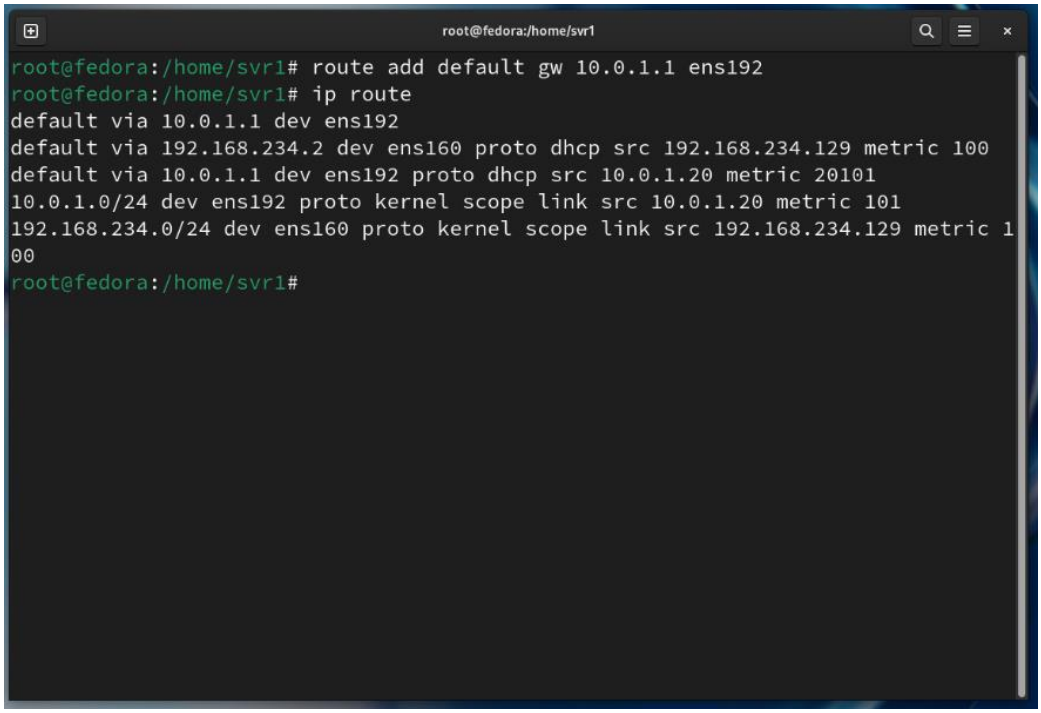
2. Change network interface status up and down, Assign the tempory IP



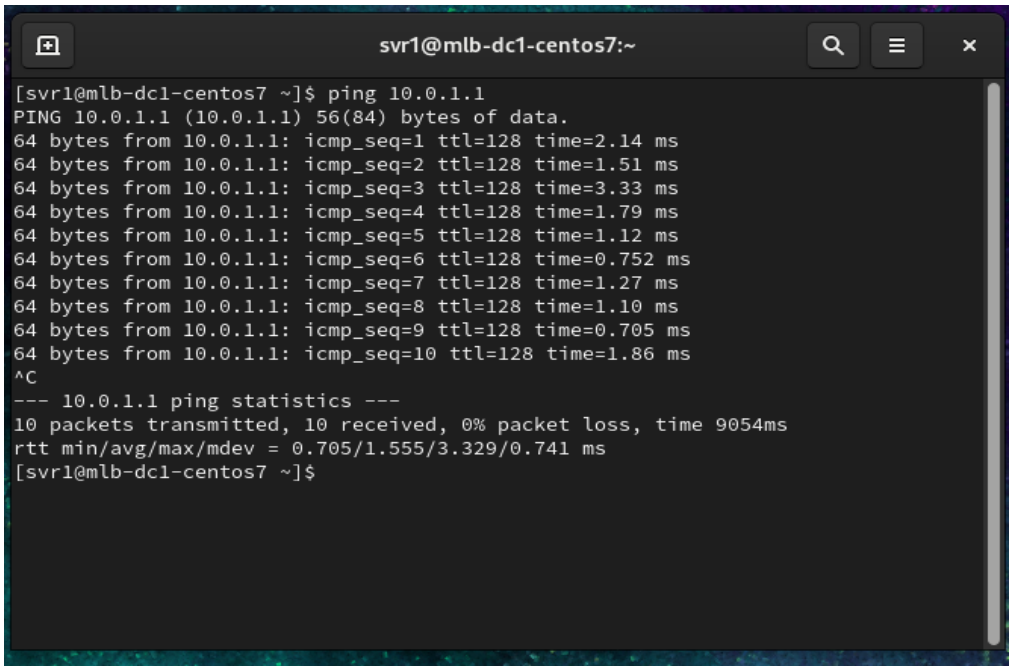


3. Assign a default gateway

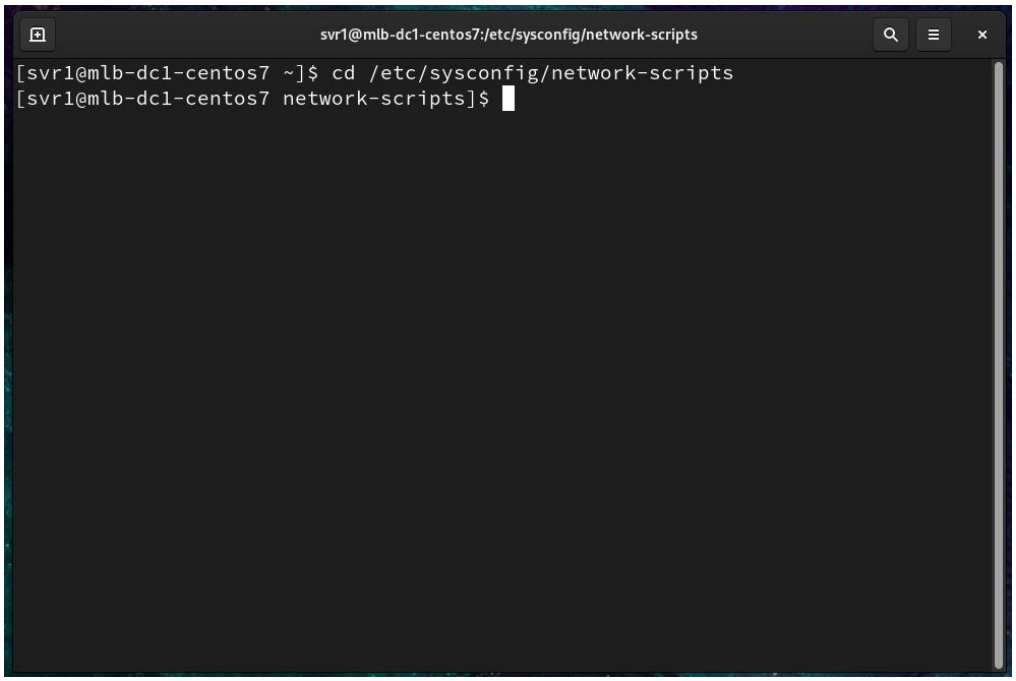




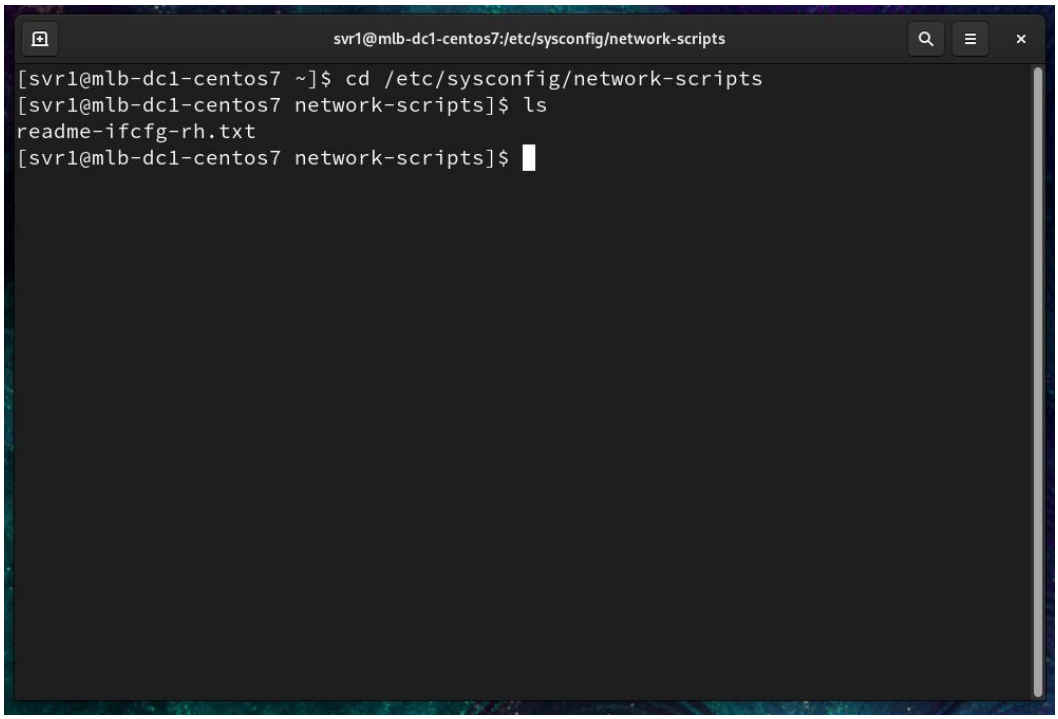
4. Go to the Server[CentOS] and check the IP address using “ping” command

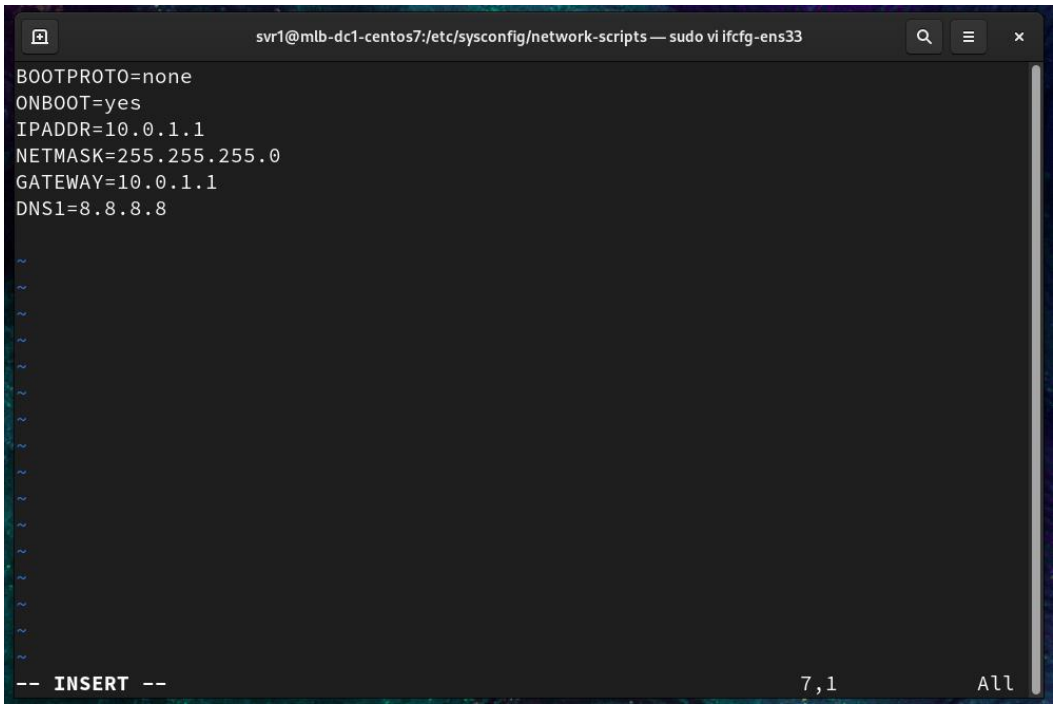


5. In the Server, Go to the below file directory using “cd” command



6. Go to the list mode using “ls” command and you can see “readme-ifcfg-rh.txt” file

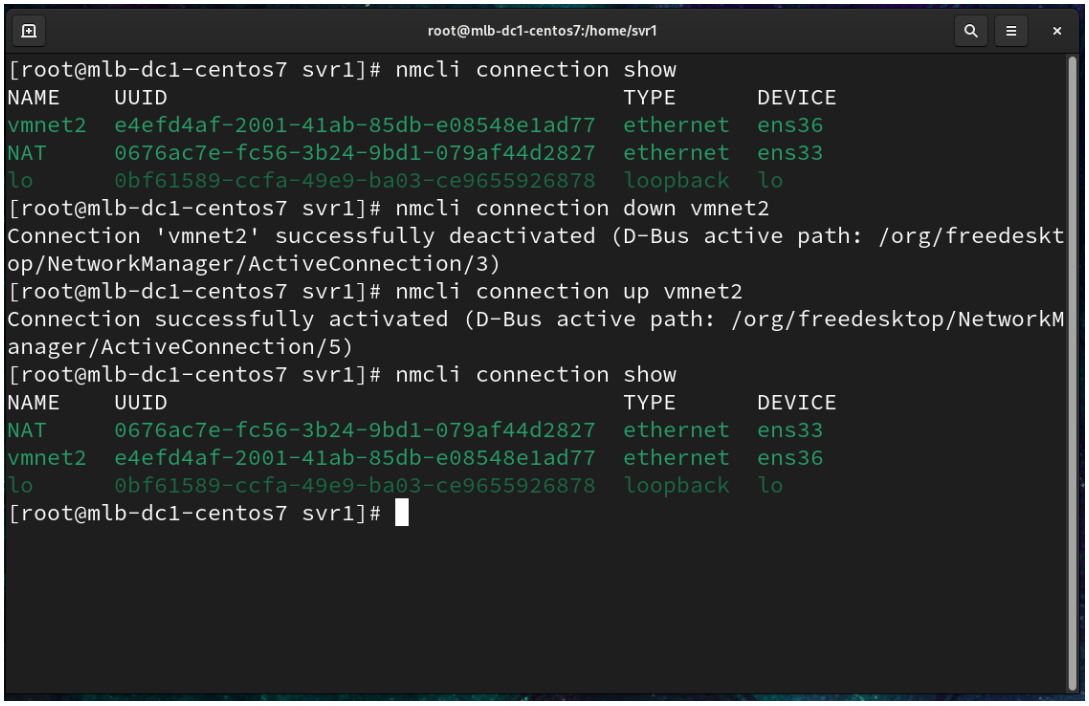




7. Then type below codes and check network interfaces

```
root@mlb-dc1-centos7:/home/svr1
[root@mlb-dc1-centos7 svr1]# nmcli connection show
NAME    UUID                                  TYPE      DEVICE
NAT     0676ac7e-fc56-3b24-9bd1-079af44d2827  ethernet  ens33
vmnet2  e4efd4af-2001-41ab-85db-e08548e1ad77  ethernet  ens36
lo      0bf61589-ccfa-49e9-ba03-ce9655926878  loopback  lo
[root@mlb-dc1-centos7 svr1]#
```

8. Verify interface name, disconnect and connect the network device again using the nmcli command

```
root@mlb-dc1-centos7:/home/svr1
[root@mlb-dc1-centos7 svr1]# nmcli connection show
NAME    UUID                                  TYPE      DEVICE
vmnet2  e4efd4af-2001-41ab-85db-e08548e1ad77  ethernet  ens36
NAT     0676ac7e-fc56-3b24-9bd1-079af44d2827  ethernet  ens33
lo      0bf61589-ccfa-49e9-ba03-ce9655926878  loopback  lo
[root@mlb-dc1-centos7 svr1]# nmcli connection down vmnet2
Connection 'vmnet2' successfully deactivated (D-Bus active path: /org/freedeskt
op/NetworkManager/ActiveConnection/3)
[root@mlb-dc1-centos7 svr1]# nmcli connection up vmnet2
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkM
anager/ActiveConnection/5)
[root@mlb-dc1-centos7 svr1]#
```

9. Double check that network interface working or not